# Birmingham geotechnical desk study

DRAFT D02 | 28 October 2026 | Original-log review revision | Public demonstration area

The six historic borehole logs now provide evidence of local fill, variable clay and granular soils, historical marl with rock bands, and weak sandstone. These are observations at the archived positions, approximately 396 to 694 m from the demonstration point. Their relevance to the actual development footprint must be established before designing new GI or selecting engineering parameters.

## Key findings from the originals

- Near-surface anthropogenic material occurs in several records: Bournbrook A records made ground to 11 ft 6 in (3.51 m); the Edgbaston holes label fill to 1.10 and 0.60 m. Bristol Road logs contain brick and other construction debris. These depths are local observations, not a site-wide fill thickness. [SP08SE16/49/50/122/123 p1]
- The older southern records describe heterogeneous clay, sand and gravel, with historical Keuper marl at Bournbrook and friable marl with rock bands at Bristol Road. The Edgbaston logs describe mixed soils above very poorly cemented sandstone beginning at 6.50 and 5.50 m. A continuous profile between these areas is unproved. [Six logs p1]
- Edgbaston drilling dates are August/September 1988, while the index says 1989. Other logs reveal differences between the index, metric annotations and original imperial depths; the two Bristol logs also contain internally conflicting total-depth notes. These conflicts remain explicit. [Six logs p1]
- Edgbaston high seating-blow entries over limited penetration are not ordinary full-penetration SPT N values. No laboratory results, energy corrections or verified design parameters were supplied in these six PDFs. [SP08SE122/123 p1]
- Groundwater remains unresolved. Bristol Road B has an old standing-water notation with an uncertain reference; Edgbaston drilling-day dry notes accompany added water. Neither establishes current conditions. The nearby EA observation remains old and Unchecked. [SP08SE50/122/123 p1; query-011]

## Revision and study basis

D02 supersedes D01 by adding visual review of all 12 pages in six original BGS PDFs, log-specific findings, a discrepancy register and a revised GI sequence. The repeated Edgbaston drawings are shared context, not six additional borehole logs. Original source files were retrieved on 28 October 2026; a factual review JSON records URLs, hashes, page references and uncertainties.

The frozen Groundwork run was generated at 19:30:45 UTC on 28 October 2026 for 52.45, -1.93, England, nominal radius 1 km. No construction boundary, surveyed site levels, layout, loads or excavation geometry has been supplied. The app export remains the unchanged historical run: 11 queries, 13 HTTP 200 responses, one unresolved coverage entry, ten open tasks and seven unknown design parameters. D02 records a subsequent review; it does not silently alter the export.

# Working ground model and site relevance

This model combines regional mapping with separate historic observations. It identifies questions for the actual works; it is not a confirmed vertical profile at the query point. Depths below refer only to the individual archived holes.

| Component | Observed evidence | Question for GI |
|---|---|---|
| Fill and disturbed ground | Made ground in Bournbrook A; labelled fill at Edgbaston; brick-bearing layers in other logs, with timber/mortar/earthenware at Bristol Road A. | Map fill extent, thickness, obstructions and variability. Assess settlement and environmental questions against actual land use and proposed works. |
| Mixed soils | Clay, silt, sand and gravel alternate in the southern records. Edgbaston BH1 includes a very-soft-described clayey sand at 1.10-1.30 m and clay between granular layers. | Identify weak pockets, granular lenses, continuity and engineering response across the works. Logged consistency/density words are historical descriptions. |
| Historical marl / rock bands | Bournbrook A: Keuper marl from 20 ft (6.10 m). Bristol A/B: marl from 8 ft 6 in / 6 ft (2.59 / 1.83 m), with deeper rock bands. | Resolve modern correlation, weathering, soil-rock transition and band hardness/thickness. No founding horizon or formation boundary is established. |
| Sand / weak sandstone | Edgbaston BH1/BH2: sandstone tops 6.50 / 5.50 m, poorly cemented and weak-described, continuing to 8.10 m. Chiselling and limited penetration recorded. | Distinguish dense sand, weathered/cemented material and rock; obtain suitable samples/core where rock behaviour affects design. |

## Regional mapped context

BGS 1:625,000 queries returned Triassic sandstone/conglomerate and mudstone/siltstone/sandstone descriptions, plus glacial sand/gravel and till/diamicton. Each query returned two of two matches, with geometry omitted. These national-scale results do not identify the exact-point unit or its thickness. Formation-equivalent fields were blank. Release metadata are bedrock 11-2-2008, version 5.17, and superficial 30-4-2003, version 1.10, not established survey dates. [query-001/002; G1]

Obtain suitably detailed mapping, georeference the historic plans and compare geological setting, elevation, location precision and engineering influence depth with the actual footprint. Do not connect the southern marl and northern sandstone records into one asserted section or transfer their contact depths to the proposed site. Current conditions may differ after decades of development.

# Original-log register and discrepancies

| Original PDF | Date on original | Index depth / distance | Original depth evidence |
|---|---|---|---|
| SP08SE14 | 1960 | 4.00 m / 670.5 m | 14 ft 6 in = 4.42 m; index differs. |
| SP08SE16 | 1959 | 15.00 m / 487.5 m | 50 ft = 15.24 m; metric annotation/index 15 m. |
| SP08SE49 | 1963 | 18.29 m / 395.5 m | Table 60 ft = 18.29 m; note including pit 58 ft 6 in = 17.83 m. |
| SP08SE50 | 20 Oct-12 Nov 1962 | 15.24 m / 439.0 m | Table 50 ft = 15.24 m; note including pit 49 ft = 14.94 m. |
| SP08SE122 | 31 Aug-1 Sep 1988 | 8.10 m / 693.5 m | Log 8.10 m; index year 1989 differs. |
| SP08SE123 | 2-5 Sep 1988 | 8.10 m / 662.4 m | Log 8.10 m; index year 1989 differs. |

All entries cite original PDF page 1. Imperial conversions use 1 ft = 0.3048 m, rounded here to 0.01 m. Original imperial entries take precedence for the stated conversion, but conflicting source statements remain unresolved. Tabulated depths describe the original log reference, with no verified present-day ground datum; Bristol pit/depth conventions require clarification.

## Quality and location checks still required

The index gives all six a position precision of plus/minus 10 m. Distances are approximate straight-line distances to the query point, not the actual site. Four index drilling years were blank; the originals supply dates above. January 2013 timestamps are index updates. Bournbrook A records 401.5 ft O.D. with a later 122.35 m annotation; exact conversion is 122.3772 m. Do not use either as a surveyed site datum. The Edgbaston page-1 ground-level fields are blank.

Seek supporting reports to resolve total-depth conventions, sample records and the Edgbaston date discrepancy. Preserve the original index values in the frozen export and record the correction/conflict separately. Terminal depths alone do not establish the reason for termination, bedrock competence or an adequate design influence depth.

## Archive coverage remains incomplete

The provider reported 423 SOBI matches in the enclosing rectangle; offsets 0 and 3 returned six, with continuation at offset 6. The six are not established as the nearest or most relevant records. The 423 is not an exact-circle count, a count of distinct physical holes or a guarantee of exhaustive archive coverage. Review of these six originals closes their acquisition/review step only. [query-003/004]

Define the site before extending bounded searches or using publisher/GIS tools. The Edgbaston shared plan identifies BH3 and TP1-TP6, which are not represented by detailed logs in these six PDFs. Acquire that report family and available laboratory results, and record relevance/exclusion decisions. An empty AGS result does not mean no investigation files exist. [query-005; SP08SE122/123 pp2-4; G2]

# Older southern logs: recorded strata

Summaries below paraphrase the original page-1 logs. Metric limits are conversions of the imperial tables, not new measurements. Detailed interval transcriptions and precision are retained in the accompanying log-review JSON.

| Record | Original depth interval | Recorded materials / observation |
|---|---|---|
| SP08SE14 1960 | 0-0 ft 9 in (0-0.23 m) | Topsoil. |
| SP08SE14 | 0 ft 9 in-7 ft (0.23-2.13 m) | Several sandy/gravelly and silty clay layers; brick traces in the upper layer. Firm clay locally. |
| SP08SE14 | 7 ft-8 ft 9 in (2.13-2.67 m) | Clayey sand with clay pockets, followed by sand/gravel with clay traces. |
| SP08SE14 | 8 ft 9 in-14 ft 6 in (2.67-4.42 m) | Brown friable silty clay, grey clayey silt, then very friable brown clayey silt. |
| SP08SE16 1959 | 0-11 ft 6 in (0-3.51 m) | Made ground. |
| SP08SE16 | 11 ft 6 in-20 ft (3.51-6.10 m) | Blackish-brown clay with sand/gravel traces. |
| SP08SE16 | 20-50 ft (6.10-15.24 m) | Historical Keuper marl description; later metric end annotation differs. |
| SP08SE49 1963 | 0-8 ft 6 in (0-2.59 m) | Brick/mortar/timber/stones to 2 ft (0.61 m), then mixed clay with brick, stone and earthenware. |
| SP08SE49 | 8 ft 6 in-18 ft 6 in (2.59-5.64 m) | Friable brown/grey marls. |
| SP08SE49 | 18 ft 6 in-60 ft (5.64-18.29 m) | Friable brown/grey marls with rock bands; table total conflicts with separate pit-inclusive note. |
| SP08SE50 1962 | 0-6 ft (0-1.83 m) | Mixed clay with brick/stones to 3 ft (0.91 m), then silty clay, pebbles, sandy pockets, brick and roots. |
| SP08SE50 | 6 ft-14 ft 3 in (1.83-4.34 m) | Friable brown/grey marls. |
| SP08SE50 | 14 ft 3 in-38 ft (4.34-11.58 m) | Friable brown/grey marls with rock bands. |
| SP08SE50 | 38-50 ft (11.58-15.24 m) | Friable brown/grey marls with very hard rock bands; conflicting total-depth note. |

No drilling method is established confidently for these four sheets. SP08SE14/16/49 show no groundwater or test results; blank fields are not dry observations. SP08SE50 records a pit, tubes withdrawn on completion and ten nominally undisturbed samples, but no laboratory results. Historical marl labels and hardness words do not establish modern formation identity, intact strength or bearing resistance. [SP08SE14/16/49/50 p1]

# Edgbaston logs: two local profiles

SP08SE122 is original BH1; SP08SE123 is BH2 at the Lucas Institute Annexe, University of Birmingham. Both used 150 mm cable percussion, with casing to 8.0 m, and end at 8.10 m. The original dates are in 1988. The log sheets give no explicit termination reason or ground elevation. [Both p1]

| SP08SE122 / BH1: m below logged ground | SP08SE123 / BH2: m below logged ground |
|---|---|
| 0-0.20: sandy topsoil labelled fill. 0.20-1.10: soft silty sandy clay fill with gravel, occasional cobbles and brick. | 0-0.20: sandy topsoil labelled fill. 0.20-0.60: soft-firm silty sandy clay fill with brick, gravel and occasional cobbles. |
| 1.10-1.30: very-soft-described clayey sand. 1.30-2.30: firm sandy clay with gravel. | 0.60-1.60: firm, becoming stiff-very stiff, sandy clay with gravel. |
| 2.30-4.00: dense gravelly coarse sand, becoming coarse sandy gravel. | 1.60-4.65: dense-very dense coarse sandy gravel, with clayey-sand pockets towards base. |
| 4.00-4.30: firm-stiff sandy clay. | No directly corresponding separate clay layer is logged; do not infer continuity from BH1. |
| 4.30-6.50: medium-dense fine sand with occasional sandstone lithorelicts. | 4.65-5.50: medium-dense fine sand, becoming very dense. |
| 6.50-8.10: very poorly cemented fine sandstone, weak, becoming moderately weak. | 5.50-8.10: very poorly cemented fine sandstone, very weak, becoming weak-moderately weak. |

All intervals above: original PDF page 1; material and consistency/density/strength descriptions are historical log observations. The parallel columns do not assert cross-hole correlation. Fill bases differ by 0.50 m and sandstone tops by 1.00 m at these two holes; neither difference can be transferred to the demonstration point.

## Shared drawings and further records

Each four-page PDF contains its individual log at page 1 and the same September 1988 drawings at pages 2-4: location plans, exploratory-hole plan and geological section (contract 1829, drawings 1-3). The plan shows BH1-BH3 and TP1-TP6, a historic proposed-building outline, retaining wall and vegetation. These are historical context, not a current site survey.

The west-east section uses questioned correlations and states that ground levels were extrapolated from another drawing. Its interpreted boulder-clay label is not a replacement for the page-1 sandy-clay descriptions. Obtain the full report and survey basis before using correlations or elevations. A groundwater-level key alone is not a measured current water level. The repeated drawings have been counted once as context, not as additional investigations. [SP08SE122/123 pp2-4]

# Testing and groundwater: usable limits

## Historic test information

Edgbaston page-1 sheets contain SPT-column values and bulk/sample markings. Unstarred reported entries range 4-77 in BH1 and 14-141 in BH2; 141 at 5.50 m is printed on BH2. These are archival entries, without supplied energy/overburden corrections, complete test sheets or a verified modern interpretation basis. No strength or density correlation is adopted here.

| Record / depth | Restricted-penetration entries | Interpretation limit |
|---|---|---|
| BH1 6.50, 7.00, 7.50, 8.00 m | 94* /150 mm; 53* /75 mm; 51* /75 mm; 64* /75 mm, respectively. | The legend defines * as SPT seating blows. These entries are not ordinary full-penetration N values. |
| BH2 6.00, 6.50, 7.00, 7.50, 8.00 m | 52*, 61*, 60*, 65*, 70*, each /75 mm, respectively. | Keep blows and penetration together. Do not pool with unstarred entries or infer competent rock from high blows. |

BH1 records one hour chiselling at 6.50-8.00 m; BH2 one hour at 5.90-8.00 m. These observations inform method selection for cemented material or hard layers. No core recovery, RQD or laboratory rock-strength results occur in these PDFs. Bristol Road B reports ten 4-inch undisturbed samples and a note extending to 50 ft, but individual sample depths, quality and laboratory results are absent. Seek the supporting records. [SP08SE50/122/123 p1]

## Water observations and their limitations

- Bristol Road B: standing-water notation 16 ft 7 in (about 5.05 m). Reference datum, measurement date/time and stabilisation are not established; do not report it as a verified depth below present ground or an AOD elevation. [SP08SE50 p1]
- Edgbaston BH1: dry on 31 August 1988, water added below 2.30 m. BH2: dry on 2 September, water added below 1.60 m; dry again on 5 September with added water. These are drilling observations with no monitoring installation or measured equilibrium water level recorded. They do not establish absence of groundwater. [SP08SE122/123 p1]
- Nearby EA Birmingham University station: latest returned value 129.14 mAOD on 16 June 2026 at 09:28:28, quality Unchecked, 134 calendar days before retrieval. The station is approximately 200 m from the point. A Qualified series label does not override reading quality. [query-009/011; G3]

mAOD is elevation, not depth. Station datum=135.23 is not a verified site ground level and cannot be subtracted to establish site groundwater depth. Aquifer relevance, perched water, pore pressures and seasonal variability remain questions for the defined works. Do not combine the 1962 notation, 1988 dry notes and 2026 station elevation into a single groundwater profile.

# Remaining gaps and closure criteria

Priorities indicate work sequence and prerequisites, not assessed hazard likelihood. Original-log review is complete for the six supplied PDFs; the frozen app task counts are unchanged. Close work packages only after a recorded technical review of the evidence below.

| Priority | Gap after original-log review | Required closure evidence |
|---|---|---|
| P0 | Site and engineering scope | Boundary, surveyed levels, proposed structures, loads, excavations, drainage and movement criteria. |
| P0 gate | Utilities, access and permissions | Asset-owner information, suitable service locating/survey, access and an agreed safe intrusive method before fieldwork. [G4] |
| P1 | Record relevance and completeness | Georeferenced historic plans; wider relevant archive coverage; full Edgbaston report including BH3/TP1-TP6; supporting Bristol sample/lab records; discrepancy decisions. |
| P1 | Site history / ground hazards | Detailed mapping, land-use and development history, walkover, relevant mining/ground-hazard information and environmental conceptual model. |
| P1-P2 | Fill and mixed-soil variability | Footprint-specific evidence of fill base, obstructions, soft pockets and granular/clay lenses, with suitable sampling and classification. |
| P2 | Weathered material / rock bands | Applicable depth coverage, weathering, sand-sandstone distinction and band/core quality where design requires; verified test quality. |
| P1-P2 | Groundwater regime | Relevant station history and project-specific monitored levels/pore pressures where needed, with surveyed datum, response zones and justified observation coverage. |
| P2-P3 | Engineering parameters | Quality-controlled field/lab evidence addressing actual works and a reasoned interpretation of applicability, uncertainty and selected values. |

## Other queries do not close these gaps

Empty AGS, landslide, EA flood-monitoring-station and current-warning results establish only that those requests returned no records. They do not prove absence of investigations, instability, flood hazard or groundwater. Review long-term flood/drainage information for the works. [query-005/008; G6]

Planning Data identified Birmingham LPA, entity 626130, reference E60000130; its representative point is not a boundary and this query did not assess all planning constraints. Brick-bearing fill warrants site-history and environmental questions but does not itself prove contamination. Mining effects, shrink-swell, faults and other relevant hazards remain subject to targeted review. [query-010; G5/G7]

# Prioritised plan: scope and desk work

## P0 Confirm the works and decision needs

Client and geotechnical designer: establish the site polygon, topographic survey, existing and proposed levels, loads, serviceability criteria, earthworks, drainage and any retained excavations. Define which foundation, settlement, excavation and groundwater decisions the GI must support. Required output: controlled site/works plan, investigation brief and constraints register.

## P1 Resolve evidence relevance before final GI design

- First georeference the six index positions and historic plans against the actual footprint and geological setting. Check the plus/minus 10 m index precision and the extrapolated Edgbaston section levels. Retain a relevance/exclusion register rather than treating proximity as proof.
- Prioritise the complete Edgbaston 1988 report family: BH3, TP1-TP6, laboratory schedules, factual report and drawing datum. Seek Bristol Road supporting project/sample records and clarification of the conflicting total-depth notes. Record the 1988/index-1989 conflict and imperial/metric differences without rewriting source evidence.
- Extend the archive search beyond the six provider-order results using bounded requests or suitable publisher/GIS tools. Target the actual site and influence area; record unsearched coverage from the 423 reported box matches. Search for usable AGS and private/client-held records where available.
- Obtain detailed geological and relevant ground-hazard information, historic maps and land-use/development records. Walk over the site to check made-ground indicators, obstructions, retaining walls, drainage, slopes, vegetation, neighbouring assets and access.
- Coordinate competent-person contamination preliminary assessment using potential source-pathway-receptor linkages. Develop chemical or gas questions from history and site evidence; construction debris alone does not establish contamination. [G5]

Required output: a site-specific conceptual ground model distinguishing observed evidence, inferred connections and unknowns; a prioritised list of engineering questions; and a preliminary GI scope. The original-log acquisition step is complete for these six PDFs, while the full-report and relevance steps remain open.

## Hold point before intrusive work

Obtain current utility information and suitable service locating/survey; confirm permissions, working space, plant/access constraints and health/environmental controls. Agree the safe method for intrusive work. HSG47 addresses planning, locating and identifying services, and safe excavation. Public borehole records provide no utility clearance. [G4]

Release criterion: designer has reviewed the works, conceptual model and investigation objectives, and responsible parties have agreed the scope and constraint controls. This demonstration point does not justify fixed hole quantities, spacing, depths, a foundation type or a GI price.

# Targeted GI, monitoring and interpretation

P2 methods below are conditional options to resolve the evidenced questions. The GI designer should select locations, quantities and depth coverage from the actual footprint, loading and engineering influence zone, and update the scope as findings emerge.

| Priority target | Field and laboratory response | Decision supported |
|---|---|---|
| Fill, obstructions and soft pockets | Use safe shallow trial pits or suitable sampling where feasible. Sample variable fill and natural soils; classify grading, moisture and plasticity, with compressibility/strength work where relevant. | Fill extent/base, earthworks reuse, variability and settlement/excavation questions. |
| Clay / sand / gravel transitions | Boreholes and suitable sampling/testing across relevant strata. Select penetration or other in-situ testing with a verified quality basis; account for gravel/cobbles and local clay lenses. | Continuity, sample quality and project-specific strength, stiffness and permeability interpretation. |
| Weathered marl / rock bands / sandstone | Choose plant capable of advancing through hard bands and weak rock. Consider rotary coring where rock behaviour governs, with recovery/discontinuity/weathering records and appropriate tests. | Distinguish weathered soil, cemented material and rock; assess rock quality and excavation/foundation response. |
| Groundwater and environmental questions | Select properly sealed monitoring installations and response zones after strata review; survey elevations and obtain repeat observations. Coordinate chemistry/gas work where the environmental model justifies it. | Relevant water levels and pore pressures, variability, excavation/drainage questions and evidenced environmental linkages. |

## Testing and monitoring scope

Schedule laboratory strength, stiffness/compressibility, permeability and chemical tests only where materials and proposed works require them, with sample quality and relevant stress/groundwater conditions specified. Historical SPT entries and seating blows cannot supply design correlations without a justified applicability and quality review. Retrieve old laboratory results where possible; new data may still be needed.

Review station/aquifer history and the possibility of local perched water. Design monitoring to address temporal variability relevant to the works, with recorded installation details, datum and quality. The current evidence does not justify a fixed monitoring duration, seasonal maximum, installation depth or dewatering rate.

## P3 Interpret, review and close

Check new factual logs, sample/test quality, locations, levels and monitoring results. Reconcile the ground model with the proposed works, identify residual uncertainty and commission supplementary investigation where needed. Record the applicability and uncertainty of derived inputs and the technical reviewer decision.

Release criterion for design: evidence and interpretation address the critical engineering questions. Friction angle, undrained strength, stiffness, permeability, design groundwater level, allowable bearing pressure and utility clearance remain unknown in D02. Closing a workflow task does not establish any of these values. Project-specific technical review is required before engineering issue.

# Sources, attribution and audit trail

Original sources: the six linked PDFs in the log register, retrieved 28 October 2026 at 20:56:17-20:56:26 UTC. All 12 pages were visually reviewed. The factual log-review JSON contains file SHA-256 hashes, byte counts, page-specific observations, converted depths and explicit uncertainty. Original scans, drawings and personal names/signatures are not redistributed in the repository.

Derived review: examples/birmingham_original_log_review_2026-10-28.json. The frozen Groundwork evidence remains examples/birmingham_report_evidence_2026-10-28.json. It retains query arguments, response dates/hashes, metadata, warnings, pagination and source terms. The following links are each query's first provider request; additional requests remain in that evidence file.

| Query | Evidence | Status | Returned | Provider request |
|---|---|---|---|---|
| query-001 | Regional geology | ok | 2 | Request |
| query-002 | Regional geology | ok | 2 | Request |
| query-003 | Historic index | partial | 3 | Request |
| query-004 | Historic index | partial | 3 | Request |
| query-005 | Historic index | empty | 0 | Request |
| query-006 | Landslide index | empty | 0 | Request |
| query-007 | EA stations | empty | 0 | Request |
| query-008 | Current warnings | empty | 0 | Request |
| query-009 | Groundwater stations | ok | 1 | Request |
| query-010 | Planning authority | ok | 1 | Request |
| query-011 | Groundwater reading | ok | 1 | Request |

## Attribution and terms

Contains British Geological Survey materials (c) UKRI 2026. The open SOBI index and 625K mapping have their stated OGL scope; linked scans/reports can carry separate underlying rights. Original records were consulted for factual summaries; access is not a grant to redistribute their contents. Landslide inventory terms include an insurance-use restriction. Contains Environment Agency information licensed under OGL v3.0.

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[G1] BGS Geology 625K. Accessed 28 October 2026.

[G2] BGS Single Onshore Borehole Index. Accessed 28 October 2026.

[G3] EA Hydrology API reference. Accessed 28 October 2026.

[G4] HSE HSG47 Avoiding danger from underground services. Accessed 28 October 2026.

[G5] EA Land Contamination Risk Management Stage 1. Accessed 28 October 2026.

[G6] GOV UK long term flood risk service. Accessed 28 October 2026.

[G7] Planning Data API documentation. Accessed 28 October 2026.

Frozen Groundwork evidence SHA-256: a8ccd4e9dd99e1e7d5c32c5b895cef50448f1be28273df6102aa45719f39f2b6. D02 is a draft for project scoping and technical review.